

Johnson C. Smith University

Mathematics Placement Test



Directions: Choose the best answer for each question and fill in your answer choice with a #2 pencil on the Scantron form provided. To ensure placement into the proper math class, guessing is discouraged, although educated guesses are appropriate. Leave questions blank if you are completely unfamiliar with a problem or have no idea how to answer it.

On the Scantron form, please provide the following information in the designated spaces:

- **Name:** Last, First, MI (exactly as you are known to JCSU)
- **Subject:** Math
- **Test No.:** If known, JCSU Identification Number. Otherwise, Birthdate (Month/Day/Year)
- **Date:** Semester and Year entering JCSU
- **Period or Hour:** Your intended major at JCSU (if you know it)

NO CALCULATORS ALLOWED

Please do not make any marks on this test packet. Additional paper will be provided for you to work out your answers.

Time limit: 50 minutes

1. $\sqrt{9+16}$

- A. 7
- B. 25
- C. 5
- D. 12.5
- E. 12

2. $27 - 18 \div 3^2 + 5 =$

- A. 30
- B. 5
- C. 6
- D. 20
- E. 29

3. $|-12 + 7|$

- A. 19
- B. 5
- C. -5
- D. $\sqrt{19}$
- E. 25

4. If $b = 7$ and $h = 12$, then $\frac{1}{2}bh =$

- A. 84
- B. 7
- C. 12
- D. 19
- E. 42

5. If $4x - 3 = 17$, then $x =$

- A. 17
- B. 3.5
- C. 16
- D. 24
- E. 5

6. $(-3x^4)(-2x^2) =$

- A. $-5x^2$
- B. $6x^8$
- C. $6x^6$
- D. $5x^6$
- E. $-6x^6$

7. Simplify $2(3x - 4y) + 12y$

- A. $3x - 4y$
- B. $3x + 4y$
- C. $6x - 8y$
- D. $6x + 8y$
- E. $6x + 4y$

8. When factored completely,
 $6x^3y^2 - 9x^2y^3 =$

- A. $3x^2y^2(2x - 3y)$
- B. $-3xy^{-1}$
- C. $3x^3y^3(2x^2 - 3y^2)$
- D. $-3xy$
- E. $3xy$

9. $\frac{x}{2} + \frac{1}{2} =$

A. $2x + 2$

B. 1

C. $\frac{x+1}{4}$

D. $\frac{x+1}{2}$

E. $x + 1$

10. If $P = 2L + 2W$, then $W =$

A. $2P + 2L$

B. $2P - 2L$

C. $P - 2L$

D. $\frac{P-2L}{2}$

E. $\frac{P}{2L}$

11. If $x = 3$ and $y = -4$, then $|x - y| =$

A. 7

B. 1

C. -1

D. -7

E. 4

12. $(2x + 1)(x - 3) =$

A. $3x - 2$

B. $2x^2 - 3$

C. $2x^2 - 5x - 3$

D. $2x^2 + 5x - 3$

E. $2x^2 + 5x + 3$

13. $\frac{3}{4} + \frac{4}{3} =$

A. 1

B. $\frac{7}{12}$

C. $\frac{12}{7}$

D. $\frac{25}{7}$

E. $\frac{25}{12}$

14. $(x + 2)^2 =$

A. $x^2 + 4$

B. $x^2 + 2x + 4$

C. $2x + 4$

D. $x^2 + 4x + 4$

E. Answer not given

15. $\frac{x}{5} \div \frac{2}{5} =$

A. $\frac{x}{2}$

B. $\frac{2x}{25}$

C. 2

D. $\frac{3x}{5}$

E. $\frac{x+2}{5}$

16. If $x \neq 3$, then $\frac{8x}{x-3} - \frac{24}{x-3} =$

A. 8

B. 3

C. $\frac{8}{x-3}$

D. $\frac{3}{x-3}$

E. $\frac{-16x}{0}$

17. $2\sqrt{3} + \sqrt{3} =$

- A. 6
- B. $3\sqrt{3}$
- C. 4.5
- D. 3
- E. 18

19. One solution of the quadratic equation $2x^2 - 5x - 3 = 0$ is

- A. 1
- B. 2
- C. 3
- D. -2
- E. -1

21. The slope of the line through the points (2,1) and (4,2) is

- A. 2
- B. 1
- C. -2
- D. $\frac{1}{2}$
- E. None of these

23. 25% of 28 is

- A. 6
- B. 7
- C. 14
- D. 16
- E. 20

18. If $x \neq 0$, then $\frac{-2x^2}{(-2x)^2} =$

- A. -1
- B. 1
- C. $\frac{1}{2}$
- D. 2
- E. -2

20. Given the formula $V = \frac{4}{3}\pi r^3$, if $V = 36\pi$, then $r =$

- A. 3
- B. 9
- C. 27
- D. 36
- E. 108

22. If x pounds of candy can be bought for 1 dollar, how many pounds of candy can be bought for y dollars?

- A. y
- B. xy
- C. $y + x$
- D. $\frac{x}{y}$
- E. $\frac{y}{x}$

24. Which quadratic equation has solutions $x = 3$ and $x = -4$?

- A. $x^2 - 7x + 12 = 0$
- B. $x^2 + 7x + 12 = 0$
- C. $x^2 + x - 12 = 0$
- D. $x^2 + x + 12 = 0$
- E. None of these

25. The length of $\overline{BC}$ in the right triangle is

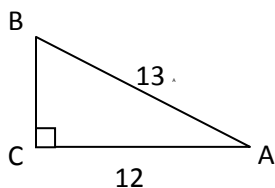
A. 5

B. 25

C. 1

D. $\sqrt{313}$

E. Cannot be determined



26. If $25 \times 25^k = 1$, then $k =$

A. 0

B. 1

C. $\frac{1}{2}$

D. $\frac{1}{25}$

E. -1

27. If the point $(2, 5)$ is on the line $y = 2x + b$, then $b =$

A. 1

B. 2

C. 3

D. 4

E. 5

28. If $2x + 1 > 7$, then

A. $x > 6$

B. $x < 6$

C. $x > 3$

D. $x < 3$

E. $x = 3$

29. The expression $\frac{3x-5}{2x+1}$ is undefined when x is equal to

A. $-\frac{1}{2}$

B. $\frac{5}{3}$

C. -1

D. $\frac{1}{2}$

E. 0

30. If $f(x) = \sqrt{169 - x^2}$, then $f(-5) =$

A. 12

B. 8

C. 13

D. 144

E. 14

31. If $f(x) = \frac{3x-1}{3}$ and $g(x) = \frac{3x+1}{3}$, then $f(g(x)) =$

A. x

B. $3x$

C. $\frac{9x^2-1}{9}$

D. $9x^2$

E. $\frac{2}{3}$

32. If $\frac{1}{2}x + 3 = 5$, then $x + 6 =$

A. 4

B. 6

C. 8

D. 10

E. 12

33. The line $5x + 12y = 60$ intersects the x -axis at P and intersects the y -axis at Q . The length of $\overline{PQ}$ is

- A. 5
- B. 7
- C. 12
- D. 13
- E. 20

35. If $x \neq 4$, then $\frac{\sqrt{x}-2}{x-4} =$

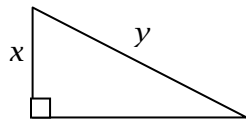
- A. $\frac{1}{\sqrt{x}+2}$
- B. $\frac{1}{\sqrt{x}-1}$
- C. $\frac{1}{x-1}$
- D. $\frac{x}{x-2}$
- E. $\frac{\sqrt{x}}{\sqrt{x}+2}$

37. For any real number x , the value of $\cos^2 x + \sin^2 x + 2$ is

- A. 2
- B. 0
- C. 3
- D. 4
- E. 1

39. The area of the right triangle pictured is

- A. $\frac{1}{2}xy$
- B. $\frac{x\sqrt{y^2-x^2}}{2}$
- C. $\frac{x\sqrt{x^2+y^2}}{2}$
- D. $\frac{x(y-x)}{2}$
- E. x^2y



34. If $2x^2 = 2x$, then

- A. $x = 2$
- B. $x = 1$ or $x = 2$
- C. $x = 0$
- D. $x = 0$ or $x = 1$
- E. $x = 1$ or $x = -1$

36. The solution of $\begin{cases} 3x + y = 7 \\ 5x - 2y = 8 \end{cases}$ is

- A. $x = 1, y = 2$
- B. $x = 2, y = 1$
- C. $x = 7, y = 8$
- D. $x = 8, y = 7$
- E. $x = 1, y = 1$

38. $\sqrt{8x^8} =$

- A. $4x^4$
- B. $2x^6$
- C. $2\sqrt{2}x^4$
- D. $8x^4$
- E. $8x^8$

40. If $\log_7 A = 3.5$ and $\log_7 B = 1$, then $\log_7 \frac{A^2}{B} =$

- A. 4.25
- B. 2.5
- C. 3.5
- D. 6
- E. 7